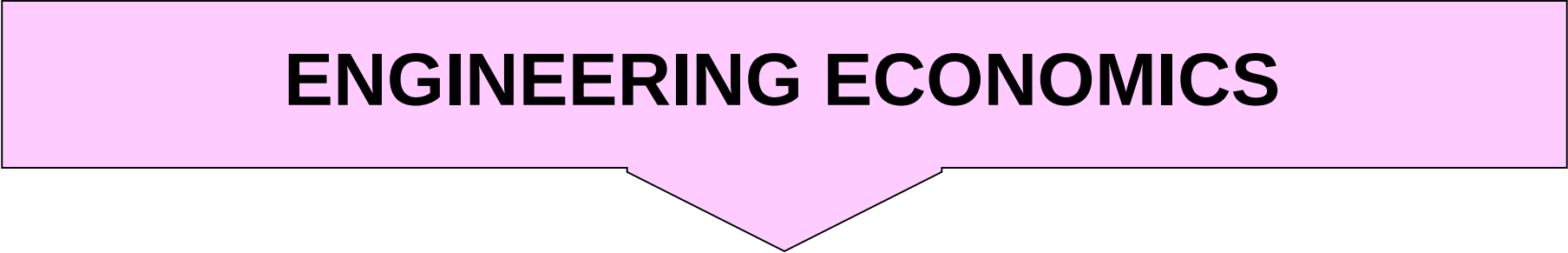


ENGINEERING ECONOMICS



LECTURE - 14



Inflation

- Persistent increase in the costs of goods and services
- Persistent decrease in buying power of dollar
- inflation rate
 - the % change in prices from one period to another
 - = $\frac{\text{price in period \#2} - \text{price in period \#1}}{\text{price in period \#1}} \times 100$

Example:

Assuming a 3% inflation rate, you would need more money each year to have the same buying power:

- Year 1: \$56,000
- Year 2: \$57,680
- Year 3: \$59,410
- Year 4: \$61,193

Your investment would need to earn more than 3% just to beat inflation.

Causes:

- 1. Cost Push**
- 2. Demand Pull**

1. COST PUSH

An increase in costs may lead to an increase in prices.

Examples:

Raw material prices (possibly from abroad) increase...
...Costs to business increase...
...Business still wants to make a profit...
...Business puts its prices up...
...Consumers can buy less with their money...
...Workers demand and receive pay increases...
...Businesses costs increase again...
...Businesses put prices up again

2. DEMAND PULL

- If there is too much demand for goods and services in the economy then prices may be forced upwards.
- Suppliers experience so much demand for their limited number of goods that they decide to put up prices

Deflation in economics is a persistent decrease in the general price level of goods and services, when inflation is below zero percent, resulting in an increase in the real value of money - a negative inflation rate.

When the inflation rate slows down (decreases, but remains positive), this is known as disinflation. It is a substantial drop in the price level.

Depression A time of economic crisis or bad times in commerce, finance, and industry, characterized by falling prices, restriction of credit, low output and investment, many bankruptcies, and a high level of unemployment (many people without jobs). A less severe crisis is usually known as a recession

Types of Capital

Capital is any form of wealth employed to produce more wealth for a firm. It has following three types.

- Fixed - used to purchase the permanent or fixed assets of the business (e.g., buildings, land, equipment, etc.)
- Working - used to support the small company's normal short-term operations (e.g., buy inventory, pay bills, wages, salaries, etc.)
- Growth - used to help the small business expand or change its primary direction.

Equity and Debt Capital

- **Equity** represents the personal investment of the owner(s) in the business.
- Does *not* have to be repaid with interest like a loan does.
- **Debt Capital** is the capital that a business raises by taking out a loan.
- Must be repaid with interest at some future date.

Bond A certificate of debt issued by a government or corporation guaranteeing payment of the original investment plus interest by a specified future date.

Credit: The giving of goods and services in return for the promise of payment at a future time. The payment usually has interest attached.

Dividend: Profits of a firm that are distributed or given out to its investors (stockholders).

Tax: A contribution for the support of a government required of persons, groups, or businesses. There are many different kinds of taxes including income, sales, state, local, federal taxes.

Tariff: A tax on imports or exports by the government.

- 1. Income taxes**
- 2. Property taxes**
- 3. Sales taxes**
- 4. Excise taxes**

Discussion